MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGG			
MID-TERM EXAMINATION/ END SEMESTER EXAMINATION		**JULY – NOV, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Automobile Engg**	**Course Type**	**PE - VII**
Course Name	**Engine Tribology**	**Course Code**	**AAE4082**
Semester	**VII**	**Max Marks**	**50**
Date	**20 – 11 - 2024**	**Duration**	**03 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A	Eluicdate friction phenonmenon and Summarize any three applications in an automobile where friction is desirable.	03	1	1,2	1	C1, C2, C13	2
1B	Materials such as Molybdenum disulphide and Graphite are used as solid lubricants, Interpret with valid reasons.	03	1	1,2	1	C1, C2, C13	2
1C	Illustrate with a neat plot, and explain the stick – slip phenomenon in friction.	04	1	1,2	1	C1, C2, C13	2
2A	Differentiate between corrosive wear and erosive wear with at least 3 differences.	04	1	1,2	1	C1, C2, C13	2
2B	Explicate the phenomenon of delamination theory of wear.	03	1	1,2	1	C1, C2, C13	2
2C	Outline at least 6 properties, a good lubricant should possess.	03	2	1,2	2	C1, C2, C13	2
3A	Interpret the drop point in a grease. With a neat sketch explain how it is determined?	04	2	1,2	2	C1, C2, C13	2
3B	With a neat sketch, elucidate how viscosity is determined using a cone on plate viscometer	03	2	1,2	2	C1, C2, C13	2
3C	List at least five different functions of a lubricant additives.	03	2	1,2	2	C1, C2, C13	2
4A	With a neat sketch explain how evaporation test of a lubricant is conducted.	03	3	1,2,3	3	C1, C2, C13	2

4B	Summarize the four conclusions drawn by Tower from his experiments conducted on journal bearing	03	3	1,2,3	3	C1, C2, C13	2
4C	A rectangular plane slider bearing with fixed shoe is operating under the following conditions: Width of the shoe = 80 mm; Length of the shoe in the direction of motion = 150 mm; Velocity of the moving member = 2 m/s; Absolute viscosity of the oil = 0.02 Pa.s; Minimum oil film thickness = 0.02 mm; Maximum oil film thickness = 0.05 mm, Find i) The load carrying capacity ii) Coefficient of friction iii) Pressure at a distance of 50 mm measured from the maximum film thickness point. Neglect end leakage.	04	3	1,2,3	3	C1, C2, C13	3
5A	A full journal bearing has the following specifications; i) Shaft diameter = 46 mm, ii) Bearing length = 66 mm, iii) Radial load = 2800 N, iv) Radial clearance to radius ratio=0.0015, v) Journal speed = 2800 rpm, vi) Viscosity of oil at operating temperature = $8.4x10^{-3}$ Pa.S. Considering the bearing as lightly loaded bearing determine, 1) Frictional force, 2) Frictional torque on Journal 3) Coefficient of friction 4) Power loss from the bearing.	04	4	1,2,3	4	C1, C2, C13	3
5B	Summarize the externally pressurized bearings stating at least 2 advantages and 2 disadvantages of the same.	03	4	1,2,3	4	C1, C2, C13	2
5C	Illustrate with a sketch and explicate the mechanism of hydrodynamic instability in journal bearings.	03	4	1,2,3	4	C1, C2, C13	3

Retain only one. Delete the others

**** Applicable for only IET-accredited courses (modules). The IET columns may be deleted for non-IET courses.***

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	17
2	13
3	10
4	10

BL	Total Marks
2	**40**
3	**10**
4	**-**
5	**-**